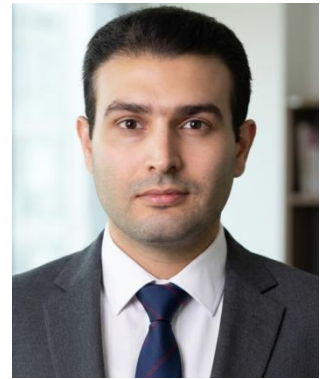


Ali Pahlevani

Robotics Software Engineer

Istanbul, Turkey | a.pahlevani2050@gmail.com | +90-5015915322

[Linkedin.com/in/ali-pahlevani](https://www.linkedin.com/in/ali-pahlevani) | [Github.com/ali-pahlevani](https://github.com/ali-pahlevani)



PROFESSIONAL SUMMARY

Robotics Software Engineer specializing in SLAM, sensor fusion, autonomous navigation, and motion planning. Develop ROS2-based robotics software in C++ and Python, with hands-on experience integrating LiDAR, cameras, IMUs, and wheel odometry. Experienced in deploying SLAM and Nav2 on autonomous mobile robots, developing simulation environments, and optimizing perception and localization pipelines for real-world applications.

WORK EXPERIENCE

Robotics Software Engineer (Freelance)

SIMOBOTICS, Bulgaria (Remote) | December 2024 – Present

- Develop and maintain a ROS2-based software stack for autonomous forklifts.
- Integrate 2D LiDAR, RGB-D cameras, IMU, and wheel encoders for sensor fusion and enhanced odometry.
- Train object detection and segmentation models for automated pallet detection and handling.
- Configure ROS2 CANopen for motor control on RPi and Industrial PCs, and develop web-based interfaces for robot control.
- Develop high-fidelity simulations and deploy SLAM solutions and Nav2 on simulation and real robots.

SLAM Engineer (Freelance)

Human Robot GmbH, Germany (Remote) | June 2025 – October 2025

- Researched and evaluated LIO-based SLAM for solid-state wide-FOV 3D LiDAR systems.
- Performed LiDAR-IMU spatiotemporal calibration for sensor fusion.
- Modified Fast-LIO2 in ROS2 for RoboSense 3D LiDAR compatibility.
- Implemented loop closure and pose graph optimization using SC-PGO.
- Generated and post-processed 3D maps using voxelization, filtering, and point cloud merging.

SLAM Engineer (Full-time)

Software Motion, China (Remote) | December 2024 – April 2025

- Conducted R&D on LiDAR-IMU SLAM systems for 3D localization and mapping.
- Improved odometry accuracy and computational efficiency using Fast-LIO2.
- Integrated solid-state and rotating 3D LiDAR with IMU for sensor fusion.
- Optimized LiDAR-IMU SLAM pipelines for real-world ADAS applications.

ADAS/SLAM Engineer (Contract)

Shoroue Innocenter, Iran | June 2020 – April 2023

- Developed SLAM, VIO, and multi-sensor fusion solutions using stereo cameras, LiDAR, IMU, and GPS.
- Designed autonomous navigation systems for indoor and outdoor mobile robots.
- Integrated path planning and motion control systems for autonomous mobile robots.

EDUCATION

M.Sc., Aerospace Engineering (Flight Dynamics and Control)

Sharif University of Technology, Iran | 2023 | GPA: 18.61 / 20

B.Sc., Aircraft Maintenance

Civil Aviation Technology College, Iran | 2021 | GPA: 15.68 / 20

TECHNICAL SKILLS

ROS2 | C++ | Python | SLAM | Sensor Fusion | Localization | Navigation | Motion Planning | CANopen | Optimization | Point Cloud Processing | Linux | Git | CI/CD | Docker | Sensor Integration | Sensor Calibration | Jetson | RPi | OpenCV | Gazebo | Isaac Sim

LANGUAGES

English – C2 (TOEFL iBT: 103) | Persian – Native | German – A1

PROJECTS

2D-Scan Merger for ROS2 | 2026

- Developed a ROS2 C++ composable node for merging multiple 2D LiDAR scans into a unified scan.

RUBI (ROS Utility Board Interface) | 2026

- Developed a single-window ROS2 interface for monitoring, controlling, and analyzing robot systems.

Dynamic World Generator | 2026

- Built a graphical wizard for designing dynamic Gazebo simulation worlds without writing SDF code.

Mobile Robot URDF Maker | 2026

- Developed a tool that automates the URDF-to-Gazebo simulation pipeline for building mobile robots with a graphical wizard.

GenZ-ICP Optimization (3D-Scan Matching) | 2025

- Optimized 3D scan matching using Eigen-based vectorization and reduced runtime by 20–50% for real-time SLAM applications.

Autonomous UGV (Isaac Sim) | 2025

- Developed a ROS2-based UGV in Isaac Sim, integrating RTAB-Map for 3D SLAM and Nav2 for autonomous navigation.

VIO-SLAM Embedded Module | 2023

- Built a Jetson-based stereo-IMU-GPS platform for embedded visual-inertial odometry and sensor-fusion experiments.

ACHIEVEMENTS

- Ranked 1st, M.Sc. Aerospace Engineering – Sharif University (2023)
- Member, National Elites Foundation (2022)
- 1st Place, Rahneshan National Autonomous Vehicle Competition (2021)